

The ACM Multimedia 2025 Grand Challenge of Truthful and Responsible Multimodal Learning

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Abstract

The Truthful and Responsible Multimodal Learning Challenge aims to foster advancements in the development of reliable and trustworthy multimodal AI systems by addressing two crucial tasks: multimodal hallucination detection and multimodal factuality detection. Task A focuses on detecting hallucinated elements in AI-generated image captions, such as fabricated objects or attributes. Task B targets verifying the factual accuracy of textual claims using visual and contextual cues. We establish benchmarks that support responsible multimodal AI in diverse real-world applications. The challenge is hosted at <https://https://mm-hall-fact.github.io/MM2025>

CCS Concepts

• Computing methodologies → Computer vision.

Keywords

Multimodal Large Language Model, Hallucination, Fact Checking

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1 Introduction

This challenge centers on key issues in multimodal Generative AI [1], targeting societal concerns such as privacy, hallucination, fairness, and bias [2, 4]. It comprises two tasks: **Multimodal Hallucination Detection** [12, 19, 20] and **Multimodal Fact Checking** [11, 13, 14], aiming to foster responsible and reliable AI.

1.1 Task Definition and Evaluation

Task A: Multimodal Hallucination Detection. Given an image and an AI-generated caption, along with a list of objects, attributes,

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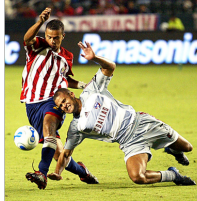
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Which team does the athlete on the right side in the below picture belong to?

Image: 

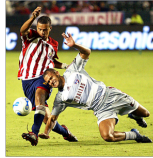
System answer: *Three players are playing the soccer ball on a blue field. The athlete on the right side, wearing the red uniform in the image, belongs to the American soccer team Club América. The scene is filled with the excitement of a soccer match.*

Which of the following systems' outputs belong to hallucination?

- ☐ A. No hallucination
☒ B. blue field
☒ C. red uniform
☐ D. soccer match

Figure 1: Multimodal Hallucination Detection task.

Claim: The athlete on the right side, wearing the white uniform in the image, belongs to the American soccer team Club América.

Image: 

Context: *The image shows two soccer players in action during a competitive match. The player on the right is wearing a white uniform with red lettering and appears to be sliding in an attempt to challenge for possession of the ball. The white uniform features "CARDINALS" on the front, suggesting it might represent a team with this name. The player on the left, wearing a red and white striped uniform with blue shorts, is trying to maintain control of the ball while balancing himself. The claim suggests that the player in white belongs to the American soccer team Club América, a well-known Mexican professional football club based in Mexico City.*

Based on the provided information, please determine whether the claim is factual.

- ☐ A. True
☒ B. False
☐ C. Partially True
☐ D. Not Verifiable

Figure 2: Multimodal Fact-Checking task.

or relationships mentioned in the caption, participants must identify which items are hallucinated—i.e., not supported by the image [3, 5, 6, 17]. The task is framed as multi-label classification.

Table 1: Data statistics for Task A: Multimodal Hallucination Detection.

Field	Train	Dev	Test
Number of images	10,000	2,000	3,000
Avg caption length	15.3 words	15.1 words	15.4 words
Hallucination types	5	5	5
GT annotations	50,000	10,000	15,000
Scene categories	18	18	18

Table 2: Data statistics for Task B: Multimodal Fact Checking.

Field	Train	Dev	Test
Number of claims	12,000	2,500	3,500
Avg context length	120 words	115 words	122 words
Number of images	12,000	2,500	3,500
Factuality labels	4	4	4
Scene categories	1	16	16

Task B: Multimodal Fact Checking Participants assess the factuality of a textual claim using three inputs: the claim, an associated image, and additional textual context. Each claim is labeled as "True", "False", "Partially True", or "Not Verifiable". This is a four-class classification problem requiring multimodal reasoning.

2 Dataset Construction

We will utilize synthetic and real-world multimodal datasets for these tasks, including balanced distributions of various scenarios.

Table 1 summarizes the statistics for Multimodal Hallucination Detection. The dataset includes 10,000 training, 2,000 development, and 3,000 test images, each paired with a caption averaging 15 words. Participants must identify five types of hallucinations (e.g., objects, attributes, relationships). The dataset contains 50,000, 10,000, and 15,000 annotated labels for the respective splits, covering 18 scene categories for diverse evaluation.

Table 2 presents the statistics for Multimodal Fact Checking. It comprises 12,000 training, 2,500 development, and 3,500 test claims, each with an image and textual context (avg. 120 words in training). Claims are classified into four labels: True, False, Partially True, and Not Verifiable. The dataset spans 16 scene categories, supporting robust evaluation across real-world scenarios.

3 Participation Summary and Conclusion

The call for participation attracted the registration of 20 teams from 4 countries, wherein 10 teams have successfully submitted results for the challenge. After two and a half months, the Top-3 teams are finalized and three of them are invited to submit a paper for the challenge. Most of their approaches utilize multimodal large models (MLLMs) [7–10, 15, 16, 18, 21] to achieve high-quality visual spatial understanding, resulting in the best performance. We hope that both the challenge data and code, as well as the systems and results of the competing teams, will serve as a valuable stepping stone for researchers and practitioners interested in the area.

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